1 (a) (i) Compton effect (ii) photoelectric effect

(b) According to de-Broglie, the motion of a moving particle is associated with a wavepacket which is formed by a superposition of two or more waves having slightly varying frequencies and wavevectors such that the velocity of the wavepacket should match with the velocity of the particle and the dimension of the maximum amplitude of the wavepacket is equal to the dimension of the particle in motion.

(c)

$$\frac{\lambda_e}{\lambda_p} = \frac{h}{p_e} \Big/ \frac{h}{p_p}$$

$$= \frac{p_p}{p_e} = \frac{\sqrt{2m_p E}}{\sqrt{2m_e E}} = \sqrt{\frac{m_p}{m_e}} = 42.83$$

(d)

$$\boxed{v_p = \frac{E}{p}} = \text{phase velocity}$$

$$\boxed{v_g = \frac{dE}{dp}} = \text{Group velocity}$$

E = Energy of the particle

p = momentum of the particle

(e)

$$V_g = \frac{dE}{dp}$$

$$E = \frac{p^2}{2m} = \text{Free particle energy only K.E.}$$

$$\therefore \frac{dE}{dp} = \frac{2p}{2m} = p/m = V_{particle}$$

(proved)

(f)

$$\lambda = \frac{hc}{E} = \frac{6.627 \times 10^{-34} \times 3 \times 10^{8}}{100 \times 1.6 \times 10^{-19}} \text{ meter}$$

$$\simeq 124 \text{ Å}$$

(2) Plank radiation Law:-

(a) The amount of radiation energy radiated from black body per unit volume with a wave length range λ & $\lambda + d\lambda$ is given by

$$E(\lambda)\,d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{\left[e^{hc/\lambda KT} - 1\right]} \quad \ldots (1)$$

(b) In deriving the radiation law Plank assumed the atoms or oscillators inside the blackbody absorb or emit energy in packets or quantized manners. E = energy of radiation of the oscillators

$$= n h\nu$$

$$n = 0, 1, 2, 3 \ldots$$

(c)

For $\lambda \to \infty$

the Plank radiation law

$$E(\lambda)\, d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{\left[e^{hc/\lambda KT} - 1\right]}$$

$\frac{hc}{\lambda KT}$ becomes small

so, $e^{hc/\lambda KT} \approx 1 + \frac{hc}{\lambda KT}$

$$E(\lambda)\, d\lambda = d\lambda \frac{(8\pi hc\ \lambda KT)}{\lambda^5 \cdot hc} = \frac{8\pi KT}{\lambda^4} d\lambda$$

(Rayleigh-Jeans Radiation Law)

For $\lambda \to 0$ (small limit)

$\frac{hc}{\lambda KT}$ is large

so $e^{hc/\lambda KT} - 1 \approx e^{hc/\lambda KT}$

So, $E(\lambda)\, d\lambda = \frac{8\pi hc}{\lambda^5} e^{-hc/\lambda KT}\, d\lambda$

(Wein's Radiation Law)

(3) (a) Differential operator linear

$$c_1 \frac{d^2 f_1(x)}{dx^2} + c_2 \frac{d^2 f_2(x)}{dx^2}$$

(b) Logarithmic operation is nonlinear in nature

(c) The strength or the magnitude of photoelectric current does not depend on frequency of incident beam.

It depends on the intensity of the incident beam.

intensity $\propto$ # of photoelectrons emitted

As # of photoelectrons increases, the magnitude of photocurrent increases.

4.

(a) $\Delta E \, \Delta t \geq \hbar/2$

Minimum conditions for uncertainty principle

$\Delta t \sim 10^{-11}$ Sec

$$\Delta E = \frac{\hbar}{2 \cdot 10^{-11}} \text{ Joule}$$

$$= 0.527 \times 10^{-34+11} \text{ Joule}$$

$$= 0.52 \times 10^{-23} \text{ Joule}$$

(b) The diameter $= 2 \times 10^{-14}$ meter

$\Delta x = 2 \times 10^{-14}$ meter

= Maximum uncertainty in measuring the diameter of the nucleus.

$\therefore$ So corresponding minimum momentum $\Delta p = \frac{h}{2 \cdot \Delta r}$

$$= 0.2635 \times 10^{-34+14}$$

$$= 0.2635 \times 10^{-20} \text{ Kg m/sec}$$

$\therefore$ Minimum K.E. of the nucleon (either proton or neutron)

$$\frac{(\Delta p)^2}{2 M_p} = \frac{(0.2635)^2 \times 10^{-40}}{2 \times 1.67 \times 10^{-27}}$$

$$= \frac{0.0694 \times 10^{-40+27}}{2 \times 1.67}$$

$$= 0.020 \times 10^{-13} \text{ Joule}$$

$$= \frac{0.020 \times 10^{-13}}{1.6 \times 10^{-19}} \text{ eV}$$

$$= 0.0125 \times 10^{6} \text{ eV}$$

As the nucleons are bound in the nucleus, the binding energy of the nucleus must be greater than the K.E.. So, binding energy should be greater than

$$- 0.0125 \times 10^{6} \text{ eV}$$